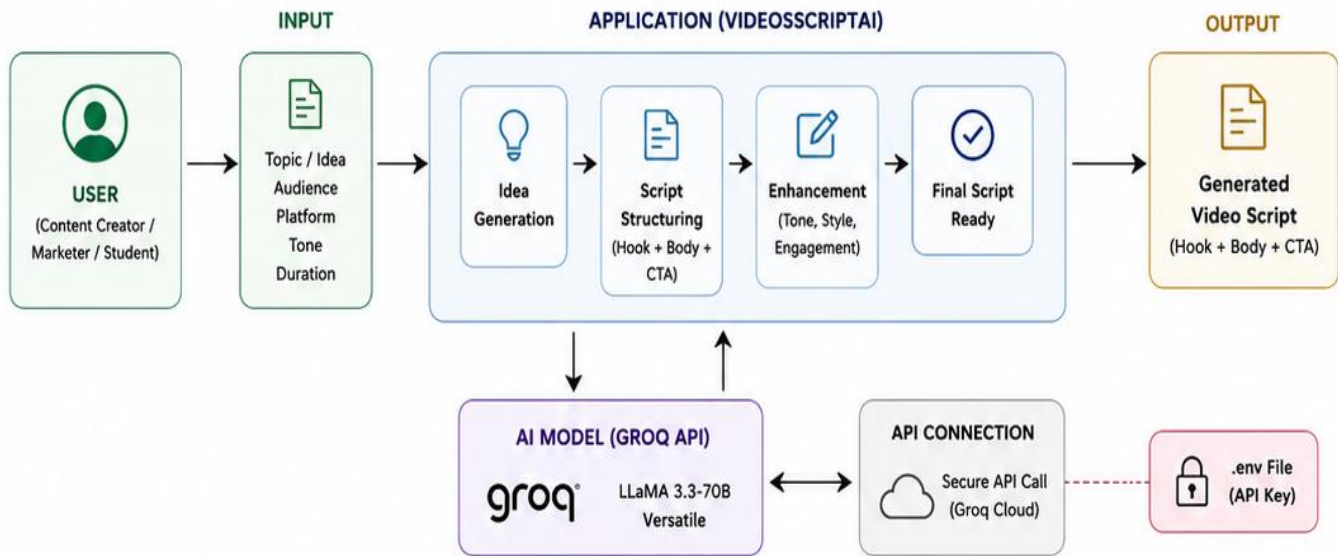


1. SYSTEM ARCHITECTURE DIAGRAM (DATA FLOW DIAGRAM)

The diagram below shows how data flows through the AI-Powered Video Script Generator system.



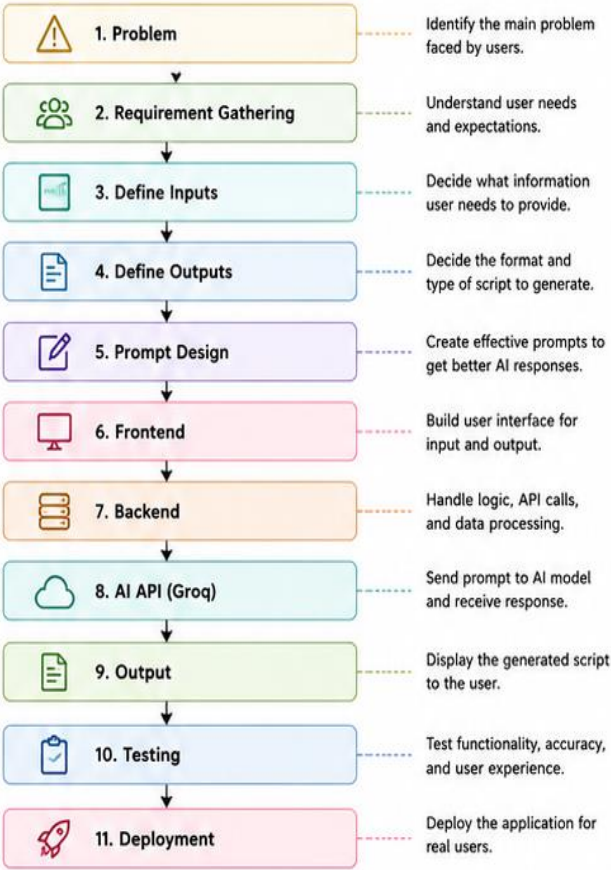
2. SYSTEM COMPONENTS (FUNCTIONAL BREAKDOWN)

Component	Example / Description
Problem	Script writing takes time and requires creativity and structure.
Users	Content creators, marketers, students, businesses.
Input	Topic / Idea, Audience, Platform, Tone, Duration.
AI Model	Groq API with LLaMA 3.3-70B Versatile model.
Prompt	Detailed instructions (prompt) to generate the script.
Output	Structured video script with Hook, Body and Call To Action (CTA).
Interface	Simple and clean web form for users to enter inputs.
API	Sends the prompt and receives the generated script from AI model.
Result	Displays the final script to the user with copy/download option.

4. TECHNOLOGY STACK & TOOLS USED

Layer	Technology / Tool
Frontend	HTML, CSS, JavaScript (or Bootstrap for UI)
Backend	Python, Flask
AI Model	Groq API – LLaMA 3.3-70B Versatile
API Handling	Python Requests Library
Environment Variables	python-dotenv (.env file for API keys)
Other	CORS (for frontend-backend communication)
Deployment	Render / Railway / Hugging Face Spaces (or any cloud platform)

3. PROJECT DEVELOPMENT PROCESS (WORKFLOW)



5. TECHNOLOGY STACK DESCRIPTION

VideoScriptAI uses Groq's API with the LLaMA 3.3-70B Versatile model to generate high-quality and creative video scripts in seconds. The backend is built using Flask (Python) which handles user requests, sends prompts to the Groq API, and returns the response. CORS is enabled to allow smooth communication between the frontend and backend. API keys are stored securely in a .env file, ensuring safety and best practices. The platform is designed to be simple, fast, and user-friendly, helping content creators, marketers and businesses save time and produce engaging video content effortlessly.

Key Benefits ★

- Saves time in script writing
- Provides creative and engaging content
- Customizable for different platforms
- Easy to use and beginner friendly
- Secure and scalable architecture